

Integrate Python and Hadoop

1. start-dfs.sh
2. start-yarn.sh
3. jps
4. hdfs dfsadmin -safemode leave → Handle Safe Mode
5. hdfs dfsadmin -safemode get
6. hdfs dfs -mkdir /forest
7. hdfs dfs -put forestfires.csv /forest → Upload Dataset to HDFS
8. hdfs dfs -ls /forest → Check
9. nano mapper.py

```
import sys
```

```
for line in sys.stdin:
```

```
    if "temp" in line:
```

```
        continue
```

```
    data = line.strip().split(",")
```

```
    if len(data) > 12:
```

```
        temp = data[8]
```

```
        print(f"{temp}\t1")
```

10. nano reducer.py

```
import sys
```

```
current_temp = None
```

```
count = 0
```

```
for line in sys.stdin:
```

```
    temp, value = line.strip().split("\t")
```

```
    if current_temp == temp:
```

```
        count += int(value)
```

```
    else:
```

```
        if current_temp:
```

```
            print(f"{current_temp}\t{count}")
```

```
        current_temp = temp
```

```
        count = int(value)
```

```
if current_temp:
```

```
    print(f"{current_temp}\t{count}")
```

11. `chmod +x mapper.py reducer.py` → Give permission

12. `hdfs dfs -rm -r /output_fire` → Run mapreduce

13.

```
hadoop jar
```

```
/home/pratik/hadoop/share/hadoop/tools/lib/hadoop-  
streaming-3.3.6.jar \
```

```
-input /forest/forestfires.csv \
```

```
-output /output_fire \
```

```
-mapper "python3 /home/pratik/mapper.py" \
```

```
-reducer "python3 /home/pratik/reducer.py" → run job
```

14. `hdfs dfs -cat /output_fire/part-00000` → view output

15. hive → start hive

16.

```
CREATE DATABASE forestdb;
```

```
USE forestdb;
```

17.

```
CREATE TABLE forestfire (
```

```
    x INT,
```

```
    y INT,
```

```
    month STRING,
```

```
    day STRING,
```

```
    FFMC FLOAT,
```

```
    DMC FLOAT,
```

```
    DC FLOAT,
```

```
    ISI FLOAT,
```

```
    temp FLOAT,
```

```
    RH FLOAT,
```

```
    wind FLOAT,
```

```
    rain FLOAT,
```

```
    area FLOAT
```

```
)
```

```
ROW FORMAT DELIMITED
```

```
FIELDS TERMINATED BY ','
```

```
TBLPROPERTIES ("skip.header.line.count"="1");
```

18.

```
LOAD DATA INPATH '/forest/forestfires.csv'
```

```
INTO TABLE forestfire;
```

19. SELECT AVG(temp) FROM forestfire;
20. SELECT MAX(temp) FROM forestfire;
21. SELECT month, COUNT(*) FROM forestfire GROUP BY
 month;
22. SELECT * FROM forestfire WHERE area > 10;